PAKO — testing guide

This guide covers the main features of PAKO for each one:

- what it is;
- where it lives (web route);
- what it needs (business type / pack, module, role, data);
- how to test it, with the expected results.

It is written for the demo data set that ships in `backend/seed/pako_demo.sql.gz`.

- Logins, roles and permission tables: **`docs/DEMO_ACCOUNTS.md`**.
- Company types and packs: **`docs/COMPANY_TYPES.md`**.

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1. Getting started

What you need

Tool	Version	Check
Docker Desktop	any recent	`docker ps`
.NET SDK	10	`dotnet --version`
Node.js + pnpm	Node 20+, pnpm 10	`pnpm --version`
Python 3	3.9+ (only to rebuild the demo data)	`python3 --version`

Start it

```
git pull origin main
pnpm install

# 1. Database. A NEW volume loads the demo data by itself on first start.
docker compose -f backend/docker-compose.yml up -d
#    Already have a local database? Replace it with the demo data (asks first, deletes
pako_dev):
scripts/demo/load_demo_db.sh

# 2. API on http://127.0.0.1:5248 (applies newer migrations on start)
dotnet run --project backend/Pako.Api/Pako.Api.csproj --launch-profile http

# 3. Web app on http://127.0.0.1:5180
pnpm --filter web dev -- --host 127.0.0.1 --port 5180
```

Open **http://127.0.0.1:5180**. Use exactly that address: the API only accepts the web app's own origin.

Logins

- **Every demo user:** password **`Demo-Passw0rd!`**.
- **Platform administrator:** `test@test.com` / `Test123!`.

Use this login	To test
`demo.admin@pako.test`	Everything at once ("Pako Demo Sh.p.k.", type General, all packs, 22 months of history)
`market.owner@pako.test`	Supermarket (Market)
`restaurant.owner@pako.test`	Restaurant
`hotel.owner@pako.test`	Hotel and events
`salon.owner@pako.test`	Services (salon)
`pharmacy.owner@pako.test`	Pharmacy
`fuel.owner@pako.test`	Fuel station
`wholesale.owner@pako.test`	Wholesale and van sales
`construction.owner@pako.test`	Construction (managed by an accounting firm)
`realestate.owner@pako.test`	Real estate sales and rentals (managed by an accounting firm)
`autoparts.owner@pako.test`	Auto parts
`firm.admin@pako.test` , `firm.accountant@pako.test`	Accounting firm with two clients
`investor@pako.test`	One user, three companies, read-only
`test@test.com`	Platform administration

All tills run in **training mode**. Coupons print "TRAINING" and nothing is sent to ATK, so you can sell, return and cancel freely.

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2. How access works (read this first)

Three layers decide what a user sees. Test them together.

1. **Business type → packs.** Each company has a type (Market, Hotel, …) that switches on its vertical packs. A pack that is off is hidden in the sidebar and its pages, and the API answers **403**. See `docs/COMPANY_TYPES.md` .
2. **Modules.** Sales, Purchases, Accounting, Reports, HR, Payroll, Inventory and POS are licensed per company by the platform administrator. Construction has no POS; Real estate has no POS or Inventory.
3. **The user's access in that company.**

Access	What it gives
Administrator / owner	everything, including users and settings
Read-only member (viewer)	sees everything, changes nothing
Firm administrator / accountant	full access to every client company of the firm (the accountant can't manage users)
Staff with a role	per area, one of: — / **View** / **Edit** (create and change drafts) / **Approve** (also post, approve, pay, cancel, refund, close a till)

The role tables are in `docs/DEMO_ACCOUNTS.md` .

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3. Common features (every company)

3.1 Company lobby, creating a company, business type

- **Where:** after login, `/select-company`; inside a company, `/settings` → Business type.
- **Needs:** any account.
  - Creating a company needs your own account (company limits apply).
  - Changing the type needs the company's administrator.

#	Steps	Expected
1	Log in as `investor@pako.test`	The lobby lists 3 companies (Market, Restaurant, Hotel). Each opens read-only.
2	Create a company, pick type **Hotel**	The new company's sidebar has Hotel and Events pages, but no Pharmacy, Forecourt or Construction.
3	In that company, open Settings → Business type and change it to **Pharmacy**	A confirmation lists the new packs. After saving, the sidebar shows Pharmacy and Hotel is gone.
4	As `market.owner`, type `/pos/hotel` in the address bar	The page says "This feature isn't used by this company".

3.2 Users, roles and permissions

- **Where:** `/admin/users`, the users and roles tabs.
- **Needs:** the company administrator, or a role with Users: Approve.

#	Steps	Expected
1	As `market.owner`, open `/admin/users`	8 users (including the investor), each with their access (administrator, role, viewer) and linked employee.
2	Open the role "Menaxher dyqani"	Shows its levels per area (POS Approve, Inventory Approve, …).
3	Invite a new user with the role "Arkëtar"	You get an invitation link (shown even without e-mail set up). Opening it and setting a password signs the user in.
4	Log in as `market.cashier`	Only POS and reference data are visible. Accounting, Payroll and Settings are not.
5	As `market.viewer`, open an invoice	It's readable, and every action button is hidden or refused.
6	As `market.employee`	Only "My HR" is visible: payslips, leave, expenses.
7	Deactivate a user, then try to log in as them	They're refused on their very next request.

3.3 Partners (customers and vendors)

- **Where:** `/customers`, `/vendors`, `/partners`, `/partner-statement`.

#	Steps	Expected
1	Create a customer with a fiscal number and payment terms	It appears in the list and can be picked on invoices.
2	Open a customer's statement for a period	Opening balance, invoices, payments, closing balance.
3	`wholesale.owner`: open "Market Viva Ferizaj"	Shows the credit limit 6,000, the overdue block (45 days) and the customer price list "Shumicë A".

3.4 Items

- **Where:** `/items`, `/item-categories`.
- **Needs:** an item needs **revenue, expense and stock accounts**; saving is disabled without them.

#	Steps	Expected
1	Create an item without accounts	The Save button stays disabled.
2	`market.owner`: open "Banane"	PLU 4011, sold by weight.
3	`market.owner`: open "Birrë Peja"	Minimum age 18 and a deposit item (the bottle).
4	`pharmacy.owner`: open "Amoksicilinë"	Pharmacy section: ATC code, prescription-only, maximum retail price 7.50.
5	`autoparts.owner`: open "Filtër vaji Bosch", tab "References & fitment"	Bosch, VW and Mann numbers; VW Golf / Škoda Octavia 2008–2013.
6	`market.owner`: open any item	No Pharmacy section and no References tab (those packs are off).

3.5 Sales: invoices, credit notes, returns, proformas

- **Where:**
  - `/invoicing`, `/invoicing/:id`;
  - `/sales-returns`, `/proforma`;
  - `/reconciliation` for payments.
- **Needs:** Sales Edit to draft; **Sales Approve to post**. Posting numbers the invoice and writes the journal entry; posted invoices can't be edited.

#	Steps	Expected
1	Draft an invoice with 2 lines and post it	It gets a number. The journal has receivable, revenue and VAT lines (Ledger). For stock items, stock goes down and cost of goods sold is posted.
2	Record a payment	The open amount goes to 0 and the invoice shows as paid.
3	Create a sales return of 1 unit from a posted invoice	Stock comes back at the original cost, and the receivable is reduced.
4	Try to return more than was sold	Refused, with the amount still returnable.
5	Invoice a customer above their credit limit	Refused with the available credit, e.g. `wholesale.owner` → "Kioska Shkolla" (limit 800).
6	As `market.cashier`, try to open `/invoicing`	Not visible or refused (no Sales access).

3.6 Purchases: bills, purchase orders, receipts

- **Where:** `/bills`, `/purchase-orders`, `/purchase-returns`, `/stock-documents` (Receipt type).
- **Needs:** Purchases Edit / Approve; Inventory Approve to post stock documents.

#	Steps	Expected
1	Create and post a bill for stock items	Stock goes up at the net cost; input VAT is posted.
2	`demo.admin`: open purchase order "EI-PO-5512"	Confirmed and received: the goods came in on a receipt.
3	Create a PO → Confirm → "Receive" → post the receipt → create the bill from the PO	The bill matches the receipt and doesn't bring the goods in twice. The order shows billed and received quantities.

3.7 Stock

- **Where:**
  - `/stock` (levels), `/stock-documents`, `/stock-reports`;
  - `/warehouses`, `/price-lists`.

#	Steps	Expected
1	Post a Transfer between two warehouses	Quantities move; no journal entry.
2	Post a Count with a shortage	Shortage moves and a journal entry to the shortage account.
3	`demo.admin`: stock of "Kos Vita"	Two lots, one expiring in 2 days. A sale takes the earliest expiry first.
4	Sell more than is in stock (the negative-stock rule is Block)	Refused, with the quantity available.
5	Post a price change (Nivelim) on a list	Records the old price, quantity and difference; no journal entry.
6	`wholesale.owner`: price list "Shumicë A", items tab	Volume tiers (a lower price from 100 pieces).

3.8 Accounting and reports

- **Where:**
  - `/ledger`, `/chart-of-accounts`, `/cost-centers`;
  - `/reports` (P&L, balance sheet, VAT declaration, ATK books, aging, cash flow, …);
  - `/year-end`, `/withholding-tax`.
- **Needs:** Accounting / Reports access. A manual journal entry asks for the user's password again.

#	Steps	Expected
1	`demo.admin`: Reports → P&L for this year	Revenue, cost of goods sold, expenses. Row links open the source document.
2	Reports → VAT declaration for last month	ATK boxes, and an Excel export.
3	Reports → checks	The 7 reconciliation checks; stock value equals the stock accounts.
4	Create a manual journal entry	Asks for the password. Posting needs debits = credits.
5	`market.accountant` (Accountant role)	Can post invoices and journal entries, but can't manage users.

3.9 HR, leave, timesheets, payroll, My HR

- **Where:**
  - `/employees`, `/departments`;
  - `/leave`, `/timesheets`, `/employee-loans`, `/commission-plans`;
  - `/payroll`, `/expense-claims`;
  - employee self-service: `/my/payslips`, `/my/leave`, `/my/expenses`, `/my/profile`.
- **Needs:** HR / Payroll roles. "My HR" needs a login linked to an employee.

#	Steps	Expected
1	`demo.admin`: `/leave`	Leave types, allocations (20 days each), 6 requests (4 approved, 2 pending).
2	Approve a pending request	The balance goes down by the working days (public holidays are skipped).
3	`/employee-loans`	An advance of €300 and a loan of €1,200, recovered from December's payroll.

#	Steps	Expected
4	Create, approve, post and pay next month's payroll	Payslips; the loan installment and the adjustments (bonus, overtime) appear. Journal and payment entries are posted.
5	Log in as `market.employee`	Only My HR. Own payslips; can request leave.
6	`hotel.hr` (HR officer)	Employees and leave yes; **no payroll or salary figures**.
7	`construction.payroll` (Payroll officer)	HR and payroll including posting and paying runs.

4. Point of sale (POS) — common to every shop

- **Where:**
  - `/pos` (stores and tills), `/pos/setup` (stores, registers, barcode rules);
  - `/pos/till/:registerId` (the till);
  - `/pos/programs` (promotions, coupons, loyalty), `/pos/cards` (gift cards, loyalty cards);
  - `/pos/customer-orders` (layaway / special orders), `/pos/reports`.
- **Needs:** POS module.
  - **POS Edit** (cashier) to sell.
  - **POS Approve** (manager) for returns, cancellations, discounts and closing a session.
- **Notes:**
  - Tills are in training mode.
  - Cash from an individual above €2,000 is refused.
  - Totals always come from the server.

#	Steps	Expected
1	`market.cashier` → `/pos` → till **DR-K2** (session open)	The till opens with the catalogue. Scan or type a barcode (e.g. `3830001020022`) to add milk.
2	Add "Banane" and enter 1.234 kg	The line shows the weight × price per kg.
3	Add "Birrë Peja"	Asks to confirm the age (18+) and adds the bottle deposit line automatically.
4	Pay cash with €20 for a smaller total	Change is calculated and the coupon prints (80 mm), marked TRAINING.
5	As the cashier, try to return an item	Refused (needs POS Approve). Log in as `market.manager`: allowed.
6	Manager closes the session: count the cash	A difference needs a reason. The session is posted as one journal entry (revenue, VAT, cash, card, cost of goods sold).
7	`demo.admin`: sell with coupon code `MIRESEVINI` (basket ≥ €20) and loyalty card `LOY-0001`	€2 off; points earned on the card.
8	Sell a gift card, then pay with it	Its balance goes down; the liability is on account 240300.
9	`/pos/reports`	Sales by item, category, day or tender, with margin; shelf labels as PDF.
10	`market.owner`: till **DR-SC** (self-checkout)	Card only, no returns. An age-restricted item asks for an attendant PIN.

5. Vertical packs — one per business type

Each section says which login to use. That company's type already has the pack on.

5.1 Market — groceries (`market.owner`)

- **Needs:** pack **grocery**; POS.
- **Data:**
  - "Supermarket Dardania — Qendra" with tills DR-K1, DR-K2 and DR-SC;
  - goods with PLUs, deposits and lots;
  - a weekend −10% on fruit, the loyalty card `DRD-0001`.
- **Test:**
  - the sales in §4;
  - the fruit promotion on Saturday or Sunday;
  - a lot that expires soon;
  - the online channel (§5.10).

5.2 Restaurant (`restaurant.owner`, `restaurant.waiter`, `restaurant.manager`)

- **Where:**
  - `/pos/floors` (floors and tables), `/pos/menu-setup` (production centres, modifiers, combos);
  - the till with profile Restaurant (**TU-K1**);
  - `/pos/kds` (kitchen screen), `/pos/usage-variance`;
  - `/pos/events` for the party hall.
- **Needs:** pack **hospitality** (events for the hall); POS. The waiter has POS Edit; voids and closing need the manager.

#	Steps	Expected
1	As `restaurant.waiter`, open TU-K1	The floor plan: Salla and Tarraca with 14 tables. 2 tables already have open tabs.
2	Open table S6, add Qebapa with "Ajvar" and 2 Coca-Cola, then Send	Tickets go to the grill and the bar. `/pos/kds` shows them live.
3	Void a sent line as the waiter	Needs the manager's PIN (`restaurant.manager`).
4	Pre-bill, then split the bill in two and pay each part	Each part is a coupon; the tab closes when everything is paid.
5	Move a tab to another table, or merge two tabs	Lines move with their state.
6	`/pos/events`	"Familja Bytyqi — synet" in 21 days. Set the final headcount to see the bill.

5.3 Hotel and events (`hotel.owner`, `hotel.reception`, `hotel.manager`)

- **Where:**
  - `/pos/hotel` (reservation chart), `/pos/hotel/setup` (room types, rooms), `/pos/hotel/guests` (guest register);
  - `/pos/events` (venues), `/pos/tickets` (ticketed events), `/pos/scan` (door scanner);
  - till **PP-REC**.
- **Needs:** packs **hotel**, **hospitality**, **events**.

#	Steps	Expected
1	`/pos/hotel`	12 rooms over the next weeks. Hans Müller and Sara Rossi are checked in today; 3 more bookings follow.
2	Open a checked-in booking	Folio: tonight's room and tourist tax per adult.
3	Check in the next arriving guest without guest details	Refused until a guest with a name and document is registered.
4	Charge a restaurant tab to a room	The amount appears on the guest's folio.
5	Check-out: pay at the till, or "Bill company" for a city-ledger invoice	The folio closes; the room becomes dirty.
6	`/pos/hotel/guests` with "foreign only"	The police register: non-Kosovo guests with document numbers.
7	`/pos/tickets` → "Mbrëmje jazz", then scan a sold ticket's QR at `/pos/scan` twice	First VALID, then ALREADY USED with the time.

5.4 Services: salon, clinic, gym, repairs (`salon.owner`, `salon.reception`)

- **Where:**
  - `/pos/appointments` (calendar), `/pos/resources` (staff, rooms, hours);
  - `/pos/memberships` (packages and memberships), `/pos/repairs`;
  - the public booking page `/book/:token` (the link is on the store in `/pos/setup`).
- **Needs:** pack **services**.

#	Steps	Expected
1	`/pos/appointments` for tomorrow	4 appointments with Teuta. Free slots respect hours and the 10-minute buffer.
2	Book an overlapping time	Refused.
3	"Check out at till" on an appointment	The till opens with the service. After payment the appointment is completed and credited to the stylist (commission).
4	Sell "Paketë 10 masazhe", then use one session at the till	Remaining 9; the session line is paid by the package.
5	Open the online booking link in a private window	Choose service, day, time; the booking appears in the calendar as Online.
6	`/pos/repairs`	Repair of "Tharëse Philips" received. Move it through the states and print the intake receipt.

5.5 Pharmacy (`pharmacy.owner`, `pharmacy.pharmacist`, `pharmacy.cashier`)

- **Where:** `/pos/pharmacy` (prescriptions, payers, controlled register) and till **BS-K1** (Prescription and Substitutes buttons).
- **Needs:** pack **pharmacy**. The controlled register needs POS Approve (the pharmacist).

#	Steps	Expected
1	At the till, add "Amoksicilinë" without a prescription	Refused: prescription-only.
2	Prescription → RX-2026-1002 (Tramadol)	Lines added with the remaining quantity. The quote shows "FSSH pays 80%" as an on-account tender; the rest is cash.
3	Dispense more than prescribed	Refused, with the quantity left.

#	Steps	Expected
4	Change a medicine's price above its maximum (e.g. Paracetamol > €3.00)	Refused at the till and in price changes.
5	Substitutes on a Paracetamol line	Panadol (same ingredient) with price and stock.
6	As `pharmacy.pharmacist`, open the controlled register for Tramadol	Every receipt and sale with prescription, patient and doctor; running balance; printable.
7	Scan a GS1 DataMatrix code (GTIN + lot + expiry)	Lot and serial land on the line; an expired pack is flagged.

5.6 Fuel station (`fuel.owner`, `fuel.attendant`)

- **Where:** `/pos/forecourt`, with tabs:
  - shift readings, dips, reconciliation;
  - tanks & nozzles, fleet cards.

The till **RG-K1** has pump/nozzle buttons and a fleet card button.

- **Needs:** pack **forecourt**.

#	Steps	Expected
1	Till: pump 1 nozzle 1, 40 litres, cash	A diesel line in litres, tied to the nozzle.
2	Fleet card **FC-2001**, 120 L of diesel	The company "Trans Kosova" is charged on account (limit €300 per fill).
3	Fleet card + a coffee	Refused: the card is fuel only.
4	Reconciliation for today	Litres pumped vs sold per nozzle (pump 1/1 is 0.4 L over); book stock vs dips per grade.
5	Enter the next shift's readings with an opening that isn't the last closing	Refused: a meter only counts up.

5.7 Wholesale and van sales (`wholesale.owner`, `wholesale.salesrep`, `wholesale.warehouse`)

- **Where:**
  - `/sales-orders` (quotes and orders);
  - `/stock-documents` (Delivery notes, type "Fletëdërgesë");
  - `/distribution/my-day`, `/distribution/routes`, `/distribution/visits`;
  - `/distribution/van-load`, `/distribution/van-settlement`;
  - `/price-lists` (customer lists, tiers).
- **Needs:** pack **distribution** for the van pages. Sales orders are in every company.

#	Steps	Expected
1	New sales order for "Market Viva Ferizaj" with 120 × Vegeta	The price comes from the customer list. From 100 pieces the tier price applies.
2	Quote → Confirm	Numbered OF-… then SO-….
3	Confirm an order that takes "Kioska Shkolla" over its €800 limit	Refused. An approver (owner) can override with a reason, which is kept on the order.
4	Order → Deliver → post the delivery note → Invoice delivered	Stock leaves on the delivery. The invoice has no second stock move; invoicing more than delivered is refused.

#	Steps	Expected
5	As `wholesale.salesrep`, open `/distribution/my-day`	The route "Ferizaj — Qendra" with 4 shops and their balances and blocks. Check in (GPS), record "Order" and the cash collected.
6	`/distribution/visits`	Visits, strike rate, drop size, collected, and a map link.
7	`/distribution/van-load` for VAN1 → Create transfer	A transfer main → van with what confirmed orders need.
8	`/distribution/van-settlement` for VAN1 today	Per item: opening, loaded, sold, returned, unloaded, closing; invoices and cash collected.

5.8 Construction (`construction.owner`, `construction.sitemanager`, `firm.accountant`)

- **Where:**
  - `/construction` (sites); `/construction/:id` with the tabs Overview, Progress bills, Costs & completion, Materials;
  - on bills: a **Project** field and a **cost code** per line.
- **Needs:** pack **construction** (no POS in this company). Approving progress bills and releasing retention need Sales Approve.

#	Steps	Expected
1	Open "Shkolla fillore Mitrovicë"	Cost code budgets and the BoQ with certified quantities.
2	Progress bills → new bill: raise 2.1 to more of its quantity → Approve → Create invoice	The amount for the period only (cumulative − previous) and 5% retention. A draft invoice tagged with the project is created.
3	Enter a cumulative quantity below the last certified	Refused.
4	Materials → issue 50 bags of cement to cost code 01	The stock leaves; the cost lands on the site.
5	Costs & completion	Actual vs budget per cost code; % complete; earned vs billed (over/under billing); retention held.
6	Log in as `firm.accountant`	Sees Ndërtimi Ilir and Arbëria Residence (the firm's clients) with full bookkeeping access.

5.9 Real estate and rentals (`realestate.owner`, `realestate.agent`)

- **Where:**
  - `/real-estate/buildings` (unit board), `/real-estate/reservations`, `/real-estate/contracts`;
  - `/rentals/leases`, `/rentals/rent-run`.
- **Needs:** packs **realEstate** and **rentals** (no POS or Inventory in this company).

#	Steps	Expected
1	Open "Kulla A"	Units coloured by state: available, reserved (A-202), sold (A-101, A-303), compensation (A-403), rented (shops).
2	Sell A-202 to someone other than the holder of the reservation	Refused. Selling to the holder converts the reservation.
3	New contract: down payment + 12 installments	The plan adds up to the price exactly.
4	Contract page: select 2 installments → Create invoice	One invoice with 2 lines. The installments show open / overdue / paid as invoices are paid.

#	Steps	Expected
5	`/rentals/rent-run` for the current month	Says "already invoiced" for the three leases (never twice). Next month creates new drafts.
6	Lease → read the electricity meter, then run next month	The rent line plus the kWh used × price. A lease that starts mid-month is charged by the day.

5.10 Online channels / webshop (Market, Pharmacy, Auto parts)

- **Where:** `/pos/online`, with tabs Orders, Cash on delivery, Channels. Webshops use the public API `/api/public/channel/{token}/...`.
- **Needs:** pack **onlineSales**.

#	Steps	Expected
1	`autoparts.owner` → Orders	"AP-WEB-1001" for pickup. Accept → Ready → "Checkout at till": the order is fiscalised and completed.
2	Channels tab	The token and the public URLs. Rotating the token makes the old one stop working.
3	`demo.admin` → Cash on delivery	Orders the courier still owes, per courier. Select them → Remit with a reference.

5.11 Auto parts (`autoparts.owner`, `autoparts.seller`)

- **Where:** `/pos/parts` (search) and the item tab "References & fitment".
- **Needs:** pack **autoParts**.

#	Steps	Expected
1	Search `03c115561h` (no spaces or dashes)	Finds the Bosch filter through the VW number.
2	Search by vehicle: Volkswagen / Golf / 2010	The filter and the brake discs.
3	Search Golf 2015	Not the filter (fitment ends in 2013).

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6. Platform administration (`test@test.com`)

- **Where:** `/platform`, with tabs for accounts, companies, account requests and the audit log.

#	Steps	Expected
1	Companies tab	Every company with its business type, packs, modules, users and activity.
2	Switch off the **hotel** pack for "Hotel Prishtina Plaza"	`hotel.owner` no longer sees the Hotel pages; the API answers 403. Switch it back on.
3	Switch off the **POS** module for a company	Its POS pages and API disappear for every user.
4	Audit log	Each change, with who, when, and before/after.
5	Set a company limit on an account and create one company too many	Refused.

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7. Permission test matrix

Log in as each user and check what's visible and what's refused. The company comes from the login.

User	Should be able to	Should NOT be able to
`market.cashier`	sell at DR-K1 / DR-K2, see items	return, cancel, close a session, open invoices, accounting, payroll, settings
`market.manager`	everything at the till, stock, prices, purchases (draft)	accounting, payroll, users
`market.stock` (Purchasing)	draft bills, items, stock documents (draft)	post stock documents (needs Approve), sales
`market.accountant`	post invoices, bills, journal entries, payroll	manage users
`market.viewer`	see everything	change anything
`market.employee`	My HR only	anything else
`restaurant.waiter`	open tabs, send to the kitchen, take payment	void sent lines, close the till
`hotel.reception`	bookings, check-in/out, guest bills	refunds, rates, closing the till
`hotel.hr`	employees, leave	payroll, salaries
`pharmacy.cashier`	sell, dispense against prescriptions	the controlled register, refunds
`wholesale.salesrep`	my day, visits, sales orders (draft)	post invoices, stock approvals
`wholesale.warehouse`	deliveries, receipts, transfers (post)	sales orders, accounting
`construction.sitemanager`	sites, materials, bills (draft)	approve progress bills, accounting
`realestate.agent`	reservations, contracts (draft)	posting invoices, rentals setup
`autoparts.accountant` (Reports only)	reports	everything else
`firm.accountant`	full bookkeeping in both firm clients	user management
`investor@pako.test`	read three companies	change anything

8. Known limits (not bugs)

- **Fiscal (ATK):** every till is in training mode. Real SEF fiscalisation needs the company's ATK certificate and registered tills.
- **Not built:** the card-terminal and fuel-pump hardware links, and the offline phone app for van sales (the server side exists).
- **Online shipping fee:** shown on the order but not on the coupon (there's no shipping item).
- **Draft invoices:** progress bills, installments and rent runs create **drafts**. Post them in Invoicing to see the ledger effect.
- **Demo dates:** "today" in the data is the day the demo set was built (see the top of `docs/DEMO_ACCOUNTS.md`). Bookings and appointments are relative to that day.

- **VAT points to confirm with an accountant:** prepaid packages, agency commission, hotel fiscalisation at check-out, new-build residential sales, rentals.

9. Reporting a bug

Please include:

1. the login and the company;
2. the page URL;
3. the steps;
4. what you expected and what happened;
5. any error message, a screenshot, and the time.

API errors are shown in the page. The API console log has the details.